Compositional Processes as Graph

Semantic Modelling and Terminology in the Anton Webern Gesamtausgabe (AWG)

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Question. How can a digital edition represent non-linear compositional processes – and keep philological claims as open as the evidence demands?

Answer: partly. `concomitates`, plain `precedes` and a certainty tag keep claims open; elsewhere openness survives only as free text and warnings.

1 One graph, two layers

Generated

- derived from the edition data
- relations read off the numbering
- op. 25: 1226 triples

Curated

- asserted by editors
- evidence, uncertainty, relations across complexes
- op. 25: 1133 triples

Draft ontology v0.1 · SPARQL in the app · Turtle export · no reasoning

2 Op. 25: the sketches cross the opus order

Edition complex op. 25 = M 317 (song I) · M 322 (song II) · M 321 (song III)

M 317 Sk4 → **M 321 Sk1–9** → **M 322 Sk1**

Curated chain: song III was sketched before song II, alongside the row table `Op25_SkRT`. The generator sees three separate islands.

3 Terms and classes

Editorial term	Class	Note
sketch (Skizze)	`Sketch` ⊂ `Inscription`	content and object at once
paratext	`Paratext` ⊂ `Inscription`	can stand in a genetic chain
row table, continuity draft	`RowTable`, `ContinuityDraft` ⊂ `Sketch`	
version (Textfassung)	`TextVersion`	abstract, not the edited sheet
work, text, sketch edition	⊂ `EditedMusicText`	`edits` its object
uncertainty	tag on a statement	no tag = certain
study composition, ink draft	not modelled yet	

4 Genetic relations

Relation	Meaning	Properties	CRM (seeAlso)
`precedes`	A begun before B, overlap allowed	transitive	P176
`concomitates`	mutual influence, order undecidable	symmetric, not transitive	P174 (overlap)
`precedesScripture`	evidence: the writing	subproperty	
`precedesContent`	evidence: the music	subproperty	

Generator: Sk1 → Sk2 as plain `precedes`, Sk2.1 `concomitates` Sk2 – never across complexes.

5 Where model and data disagree

Model lags data

- staves, measures as free text (“1c–3c”)
- writing instruments missing

Data lag model

- sketchbook 3 described twice
- 29 sketches as Sk## placeholders
- datings without year

12 generator warnings: `emptyItem` `folioConflict` `labelConflict` `incompleteDate` `noComplex` `noDocument` `noFolio` `noParentSketch` `placeholder` `unknownConvolute` `unknownItem` `unresolvedReference`

6 Standards: alignments, not the model

Standard	Mapping	Link
LRMoo	complex ≈ F1 · version, sketch ≈ F2 · document ≈ F5	`seeAlso`
CIDOC CRM	`precedes` ≈ P176 · `concomitates` ≈ P174 (via F28)	`seeAlso`
CRMinf	editorial statement ≈ I2 Belief	`seeAlso`
DC terms, SKOS, PROV	parts · concept schemes · attribution	used directly
MEI	not integrated (sheets are SVG)	–

7 Next steps

- named graphs for ink and pencil layers
- statement nodes only where uncertainty or a justification needs them
- persons and archives

For discussion. Whose genetic order does the graph show – Webern’s or ours?
How much of Webern’s compositional process survives translation into shared standards?